

1 L^AT_EX Template for Journal article

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3 **Abstract**

4 This is the abstract.

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6 1. Introduction

7 Try to say something [?].

8 1. error analysis

9 2. Reissner–Mindlin plate formulation

10 2.1. Governing equations for Mindlin plate

11 Consider a thick plate $\Omega \times [-\frac{t}{2}, \frac{t}{2}]$ with its mid-surface Ω and thickness t in
 12 a Cartesian coordinate system, as shown in Figure, where x_α , $\alpha = 1, 2$ are the
 13 in-plane coordinates on Ω and x_3 is the coordinate for the thickness direction.
 14 In accordance with Reissner-Mindlin hypothesis and inconsideration of the in-
 15 plane displacement of Ω , the displacement u_i , $i = 1, 2, 3$ are given as follows:

$$\begin{cases} u_\alpha(\mathbf{x}) = -x_3\varphi_\alpha(x_1, x_2) \\ u_3(\mathbf{x}) = w(x_1, x_2) \end{cases}, \quad \alpha = 1, 2 \quad (1)$$

16 where w stands for the deflection and φ_α presents α th direction component for
 17 rotation $\boldsymbol{\varphi}$. The corresponding infinitesimal strain components ε_{ij} are listed as
 18 follows:

$$\begin{cases} \varepsilon_{\alpha\beta} = \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}) = x_3\kappa_{\alpha\beta} \\ \varepsilon_{\alpha 3} = \frac{1}{2}\gamma_\alpha \\ \varepsilon_{33} = 0 \end{cases} \quad (2)$$

19 in which $\kappa_{\alpha\beta}$ and γ_α are the components for the in-plane curvature $\boldsymbol{\kappa}$ and shear
 20 strain $\boldsymbol{\gamma}$ respectively, the $\boldsymbol{\kappa}$ and $\boldsymbol{\gamma}$ are defined as:

$$\boldsymbol{\kappa} = -\frac{1}{2}(\nabla\boldsymbol{\varphi} + \boldsymbol{\varphi}\nabla) \quad (3)$$

$$\boldsymbol{\gamma} = \nabla w - \boldsymbol{\varphi} \quad (4)$$

21 According to the isotropic Mindlin plate theory [1], the stress components
 22 σ_{ij} are given as follows:

$$\begin{cases} \sigma_{\alpha\beta} = D_{\alpha\beta\gamma\eta}\varepsilon_{\gamma\eta} \\ \sigma_{\alpha 3} = G\gamma_\alpha \\ \sigma_{33} = 0 \end{cases} \quad (5)$$

23 where $D_{\alpha\beta\gamma\eta}$ is denoted as the component for the fourth-order in-plane consti-
 24 tutive tensor $\mathbf{D}$ and G is shear modulus, and they can be evaluated by Young's
 25 modulus E , Poisson's ratio ν as:

$$\mathbf{D} = \frac{E}{1-\nu^2} \left(\nu \mathbf{1} \otimes \mathbf{1} + \frac{1}{2}(1-\nu)\mathbf{I} \right) \quad (6)$$

$$G = \frac{E}{2(1+\nu)} \quad (7)$$

in which $\mathbf{1} = \delta_{\alpha\beta} \mathbf{e}_i \otimes \mathbf{e}_j$ and $\mathbf{I} = \delta_{\alpha\gamma} \delta_{\beta\eta} + \delta_{\alpha\eta} \delta_{\beta\gamma} \mathbf{e}_\alpha \otimes \mathbf{e}_\beta \otimes \mathbf{e}_\gamma \otimes \mathbf{e}_\eta$ represent the second- and fourth-order identity tensors, and $\delta_{\alpha\beta}$ is Kronecker delta function. Further integrating the stress components through the thickness direction, we can obtain the moment resultant $\mathbf{m}$ and shear force resultant $\mathbf{q}$, given as follows:

$$\mathbf{m} = \int_{-t/2}^{t/2} \sigma_{\alpha\beta} dx_3 \mathbf{e}_\alpha \otimes \mathbf{e}_\beta = \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} \quad (8)$$

$$\mathbf{q} = k \int_{-t/2}^{t/2} \sigma_{\alpha 3} dx_3 \mathbf{e}_\alpha = k G t \boldsymbol{\gamma} \quad (9)$$

where k presents the shear correction factor that is chosen as $k = \frac{4}{5}$ in this work. It should be noted that, the geometric parameter for the moment resultant has two more order power of thickness than shear force resultant. As the thickness reducing, the parameter for the moment resultant will much less than that for shear force resultant, for a specific external force, leading to a constraint of shear strain trending to be zero, i.e. $t \rightarrow 0, \boldsymbol{\gamma} \rightarrow \mathbf{0}$.

Consequently, the governing equations for Reissner-Mindlin plate can be expressed as follows:

$$\begin{cases} \nabla \cdot \mathbf{m} - \mathbf{q} = \mathbf{0} & \text{in } \Omega \\ \nabla \cdot \mathbf{q} + \bar{p} = 0 & \text{in } \Omega \\ \mathbf{q} \cdot \mathbf{n} = \bar{q}_n & \text{on } \Gamma_q \\ \mathbf{m} \cdot \mathbf{n} = \bar{\mathbf{m}}_n & \text{on } \Gamma_m \\ \varphi = \bar{\varphi} & \text{on } \Gamma_\varphi \\ w = \bar{w} & \text{on } \Gamma_w \end{cases} \quad (10)$$

where $\bar{p}$ is the transverse load, $\bar{q}_n$ and $\bar{\mathbf{m}}$ are the prescribed shear force and moment on the natural boundary Γ_q and Γ_m respectively, $\bar{w}$ and $\bar{\varphi}$ are the prescribed deflection and rotation on the essential boundary Γ_w and Γ_φ respectively, and $\mathbf{n}$ is the unit outward normal vector on the boundary. All the boundary segments satisfy the following relationship:

$$\Gamma_q \cup \Gamma_m = \Gamma_w \cup \Gamma_\varphi = \Gamma, \quad \Gamma_w \cap \Gamma_\varphi = \Gamma_q \cap \Gamma_m = \emptyset \quad (11)$$

and Γ is the boundary of Ω .

2.2. Mixed-formulation

The thick plate Galerkin formulation generally use the three-variable weak form to immigrate the locking issue. To obtain the corresponding Galerkin weak form, the complementary energy functional is given by:

$$\begin{aligned} \Pi(\mathbf{u}, \mathbf{q}) = & \int_{\Omega} \frac{t^3}{24} \boldsymbol{\kappa} : \mathbf{D} : \boldsymbol{\kappa} d\Omega - \int_{\Omega} \frac{1}{2kGt} \mathbf{q} \cdot \mathbf{q} d\Omega - \int_{\Omega} \varphi \cdot \mathbf{q} d\Omega \\ & + \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma + \int_{\Gamma_m} \varphi \cdot \bar{\mathbf{m}}_n d\Gamma + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{q}_n) d\Gamma \\ & - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega \end{aligned} \quad (12)$$

48 where $\mathbf{u}$ stands for a displacement set containing deflection w and rotation $\boldsymbol{\varphi}$,
 49 i.e. $\mathbf{u} = \{w, \boldsymbol{\varphi}\}$. Introducing the standard variational operation herein leads to
 50 the following Galerkin weak form: find $w \in W$, $\boldsymbol{\varphi} \in \Phi$, $\mathbf{q} \in Q$,

$$\begin{aligned} a_\varphi(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) + b_\varphi(\delta\boldsymbol{\varphi}, \mathbf{q}) &= f_\varphi(\delta\boldsymbol{\varphi}), \quad \forall \delta\boldsymbol{\varphi} \in \Phi \\ b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\ a_q(\delta\mathbf{q}, \mathbf{q}) + b_q(\delta\mathbf{q}, \mathbf{u}) &= f_q(\delta\mathbf{q}), \quad \forall \delta\mathbf{q} \in Q \end{aligned} \quad (13)$$

51 where the deflection space W , rotation space Φ and shear force space Q are
 52 defined as:

$$W := \{\} \quad (14)$$

$$\Phi := \{\} \quad (15)$$

$$Q := \{\} \quad (16)$$

53 and the bilinear forms and linear forms in Eq. (13) are detailed as follows:

$$a_\varphi(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) = \int_{\Omega} \delta\boldsymbol{\kappa} : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} d\Omega \quad (17)$$

$$a_q(\delta\mathbf{q}, \mathbf{q}) = - \int_{\Omega} \frac{1}{2kGt} \delta\mathbf{q} \cdot \mathbf{q} d\Omega \quad (18)$$

$$b_q(\mathbf{q}, \mathbf{u}) = b_w(w, \mathbf{q}) + b_\varphi(\boldsymbol{\varphi}, \mathbf{q}) \quad (19)$$

$$b_\varphi(\boldsymbol{\varphi}, \mathbf{q}) = - \int_{\Omega} \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega \quad (20)$$

$$\begin{aligned} b_w(w, \mathbf{q}) &= \int_{\Gamma_q} w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} w \nabla \cdot \mathbf{q} d\Omega \\ &= \int_{\Omega} \nabla w \cdot \mathbf{q} d\Omega - \int_{\Gamma_w} w \mathbf{q} \cdot \mathbf{n} d\Gamma \end{aligned} \quad (21)$$

$$f_\varphi(\boldsymbol{\varphi}) = - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \quad (22)$$

$$f_w(w) = \int_{\Gamma_q} w \bar{q}_n d\Gamma + \int_{\Omega} w \bar{p} d\Omega \quad (23)$$

$$f_q(\mathbf{q}) = - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \quad (24)$$

54 Furthermore, we introduce the approximation schemes to discrete these three
 55 variables, namely w_h , $\boldsymbol{\varphi}_h$, $\mathbf{q}_h$, these approximants are represented by shape
 56 functions N_I^w , N_K^φ , N_M^q and their corresponding nodal coefficients w_I , $\boldsymbol{\varphi}_K$, $\mathbf{q}_M$

57 respectively, yields:

$$\begin{cases} w_h(\mathbf{x}) = \sum_{I=1}^{n_w} N_I^w(\mathbf{x}) w_I \\ \varphi_h(\mathbf{x}) = \sum_{K=1}^{n_\varphi} N_K^\varphi \varphi_K \\ \mathbf{q}_h(\mathbf{x}) = \sum_{R=1}^{n_q} N_R^q(\mathbf{x}) \mathbf{q}_R \end{cases} \quad (25)$$

58 Substituting Eq. (25) back into Eq. (13) can obtain the Galerkin weak formu-
59 lation: find $w_h \in W_h$, $\varphi_h \in \Phi_h$, $\mathbf{q}_h \in Q_h$,

$$a(\delta\varphi_h, \varphi_h) + b_\varphi(\delta\varphi_h, \mathbf{q}_h) = f_\varphi(\delta\varphi_h), \quad \forall \delta\varphi_h \in \Phi_h \quad (26a)$$

$$b_w(\delta w_h, \mathbf{q}_h) = f_w(\delta w_h), \quad \forall \delta w_h \in W_h \quad (26b)$$

$$b(\delta\mathbf{q}_h, \mathbf{u}_h) = f_q(\delta\mathbf{q}_h), \quad \forall \delta\mathbf{q}_h \in Q_h \quad (26c)$$

60 where $\mathbf{u}_h = \{w_h, \varphi_h\}^T$. The related approximation spaces are defined as:

$$W_h := \text{span}\{N_I^w\}_{I=1}^{n_w} \quad (27)$$

$$\Phi_h := \text{span}\{N_K^\varphi\}_{K=1}^{n_\varphi} \quad (28)$$

$$Q_h := \text{span}\{N_R^q\}_{R=1}^{n_q} \quad (29)$$

61 FIXME: approximation types

$$\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} & \mathbf{K}^{q\varphi T} \\ \mathbf{0} & \mathbf{0} & \mathbf{K}^{qwT} \\ \mathbf{K}^{q\varphi} & \mathbf{K}^{qw} & \mathbf{K}^{qq} \end{bmatrix} \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \\ \mathbf{d}^q \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi \\ \mathbf{f}^w \\ \mathbf{f}^q \end{Bmatrix} \quad (30)$$

$$\left(\underbrace{\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{K}^{\text{bending}}} + \underbrace{\begin{bmatrix} \tilde{\mathbf{K}}^{\varphi\varphi} & \tilde{\mathbf{K}}^{\varphi w} \\ \tilde{\mathbf{K}}^{w\varphi} & \tilde{\mathbf{K}}^{ww} \end{bmatrix}}_{\tilde{\mathbf{K}}^{\text{shearing}}} \right) \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi - \tilde{\mathbf{f}}^\varphi \\ \mathbf{f}^w - \tilde{\mathbf{f}}^w \end{Bmatrix} \quad (31)$$

62

$$\tilde{\mathbf{K}}^{\varphi\varphi} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{q\varphi} \quad (32)$$

$$\tilde{\mathbf{K}}^{\varphi w} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (33)$$

$$\tilde{\mathbf{K}}^{ww} = -\mathbf{K}^{qwT} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (34)$$

63 **Remark 1.** As shown in Eq. (26), there are two constraints in thick plate
64 mixed formulation. One, as listed in Eq. (26b), is a constraint mainly regarding
65 the shear force resultant on natural boundary Γ_q and its divergence in entire
66 domain. The other is deflection essential boundary condition Γ_w and the shear
67 strain enforcement in whole domain, as listed in Eq. (26c). However, for the
68 most of pioneer works, only consider the shear strain constraint, and that is not
69 sufficient for a satisfaction result.

70 **Remark 2.**

71 3. Optimal constraint ratio for shear locking

72 The previous section has been mentioned that the shear locking is consisted
 73 of two parts, shear force locking and shear strain locking. The corresponding
 74 LBB conditions are listed as follows:

- 75 • Shear force locking LBB condition:

$$\inf_{w_h \in W_h} \sup_{\mathbf{q}_h \in Q_h} \frac{|b_w(w_h, \mathbf{q}_h)|}{\|w_h\|_W \|\mathbf{q}_h\|_Q} \geq \beta_q \quad (35)$$

- 76 • Shear strain locking LBB condition:

$$\inf_{\mathbf{q}_h \in Q_h} \sup_{\mathbf{u}_h \in W_h \times \Phi_h} \frac{|b(\mathbf{q}_h, \mathbf{u}_h)|}{\|\mathbf{q}_h\|_Q \|\mathbf{u}_h\|_{W \times \Phi}} \geq \beta_\gamma \quad (36)$$

77 where, β_q and β_γ are LBB coefficients. The norms in the above LBB conditions
 78 are defined as follows:

$$\|w_h\|_W^2 = \int_{\Omega} \delta w_h k G h w_h d\Omega \quad (37)$$

$$\|\mathbf{q}_h\|_Q^2 = \int_{\Omega} \frac{\delta \mathbf{q}_h \cdot \mathbf{q}_h}{k G t} d\Omega \quad (38)$$

$$\|\boldsymbol{\varphi}_h\|_{\Phi}^2 = \int_{\Omega} \delta \boldsymbol{\kappa}_h : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa}_h d\Omega \quad (39)$$

$$\|\mathbf{u}_h\|_{W \times \Phi}^2 = \|w_h\|_W^2 + \|\boldsymbol{\varphi}_h\|_{\Phi}^2 \quad (40)$$

79 Next, we will focus on these two LBB conditions respectively and employ the
 80 complete-polynomial-based estimate method [2] to investigate the relationship
 81 between their LBB coefficients β_q and β_γ and the degree of freedoms (DOFs)
 82 for variables w_h , $\boldsymbol{\varphi}_h$ and $\mathbf{q}_h$. The detailed path for the complete-polynomial-
 83 based estimate method is listed in Appendix A. Consequently, the optimal ratio
 84 of degrees of freedoms (DOFs) between variables can be revealed based on these
 85 two estimators.

86 3.1. Estimator for β_q

87 The coefficient β_q controls the behavior of shear force locking in Eq. (26b),
 88 the detailed enforcement is listed as follows:

$$\begin{aligned} & b_w(\delta w_h, \mathbf{q}_h) - f_w(\delta w_h) = (\delta w_h, \mathcal{W} \mathbf{q}_h) - f_w(\delta w_h) \\ & = \int_{\Gamma_q} \delta w_h (\mathbf{q}_h \cdot \mathbf{n} - \bar{q}_n) d\Gamma - \int_{\Omega} \delta w_h (\nabla \cdot \mathbf{q}_h + \bar{p}) d\Omega \\ & = 0 \end{aligned} \quad (41)$$

89 where, the constraints includes two parts, one is the equilibrium equation be-
 90 tween shear force resultant and transverse load in Ω , the other is the natrual

91 shear force boundary condition on Γ_q . In accordance with the general form of
 92 completed-polynomial-based estimator mentioned in Appendix A, the operator
 93 $\mathcal{W} : Q \rightarrow W$ is a specific instance of $\mathcal{P}$ in Theorem.1, and is defined as:

$$\mathcal{W} = \begin{cases} \mathcal{W}^\Omega = -\nabla \cdot & \text{in } \Omega \\ \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}, \mathcal{W}^{\Gamma_q} = \mathbf{n} \cdot & \text{on } \Gamma_q \end{cases} \quad (42)$$

94 where, based on the trace theorem [], $\mathcal{T}^{\Gamma_q} : Q \rightarrow Q|_{\Gamma_q}$ is a trace operator map-
 95 ping the shear force $\mathbf{q}$ from domain Ω to the boundary Γ_q , while the parametric
 96 variable of $\mathbf{q}$ on boundary can be represented using s , i.e. $\mathbf{q}(s) \in Q|_{\Gamma_q}$. Ac-
 97 cording to Theorem.1, the completed-polynomial-based estimator of β_q has the
 98 following form:

$$\beta_q \leq \inf_{Q' \subset Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q} \sup_{\mathbf{q} \in Q'} \frac{\|\mathcal{W}\mathbf{q}\|_W}{\|\mathbf{q}\|_Q} + O(h) \quad (43)$$

99 in which $\mathcal{I}_h^q : Q_{n_q} \rightarrow Q_h$ is the interpolation operator for shear force resultant.
 100 $\mathcal{W}_h : Q_h \rightarrow W_h$ is the orthogonal operator of $\mathcal{W}$, i.e. $(\delta w_h, \mathcal{W}\mathbf{q}_h - \mathcal{W}_h\mathbf{q}_h) = 0$.
 101 $Q_{n_q} \subset Q$ is the polynomial space that has the identical dimensions with Q_h , i.e.
 102 $\dim Q_{n_q} = \dim Q_h = 2n_q$.

103 From Eq. (43), the LBB condition of Eq. (35) can be maintained while the
 104 first term on right-hand side of Eq. (43) is non-zero, which should be necessarily
 105 required that:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \quad (44)$$

106 where, obviously, leading the definitions of $\mathcal{I}_h^q$ and Q_{n_q} , and the rank-nullity
 107 theorem, the left-hand side of Eq. (44) can be rewritten as:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) &= \dim(Q_h \setminus \ker \mathcal{W}_h) \\ &= \dim Q_h - \dim(\ker \mathcal{W}_h) \\ &= \dim(\text{Im} \mathcal{W}_h) \\ &= \dim W_h = n_w \end{aligned} \quad (45)$$

108 In contrast, to evaluate the right-hand side of Eq. (44), we first introduce
 109 the definition of operator $\mathcal{W}$ in (42), $\ker \mathcal{W}$ can be represented as:

$$\ker \mathcal{W} = \ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q} \quad (46)$$

110 According to above equation, the space $Q_{n_q} \setminus \ker \mathcal{W}$ can be expressed by two
 111 terms, one is regarding the kernel space of $\mathcal{W}^\Omega$, the other is regarding the kernel
 112 space of $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$, yields:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}) &= \dim(Q_{n_q} \setminus (\ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cup \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) + \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &\quad - \dim((Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cap (Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \end{aligned} \quad (47)$$

where the second and third equations of Eq. (47) is derived based on De Morgan's theorem[] and Grassmann's theorem[] respectively. For clarity, these two theorems are illustrated in Appendix B.

For the first term regarding the kernel space of $\mathcal{W}^\Omega$, in accordance with rank-nullity theorem[3], the dimension of this space is equal to the that of the image space of $\mathcal{W}^\Omega$ as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) = \dim Q_{n_q} - \dim(\ker \mathcal{W}^\Omega) = \dim(\text{Im} \mathcal{W}^\Omega) \quad (48)$$

Because of the Q_{n_q} is the second order polynomial tensor space with dimension of $2n_q$ and the $\mathcal{W}^\Omega$ is the divergence operator acting on Q_{n_q} , the polynomial in $\text{Im} \mathcal{W}^\Omega$ is one order lower than polynomial order of Q_{n_q} . The following equation presents the detailed monomial basis of Q_{n_q} , and the kernel space of operator $\mathcal{W}^\Omega$:

$$Q_{n_q} = \text{span} \left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix}, \begin{pmatrix} x \\ -y \end{pmatrix}, \dots, \begin{pmatrix} x \\ y \end{pmatrix}, \dots}_{\ker \mathcal{W}^\Omega}, \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}, \dots}_{Q_{n_q} \setminus \ker \mathcal{W}^\Omega} \right\} \quad (49)$$

To simplify the expression, we define two operators to implement the transformation between the order of a complete polynomial space, namely n_o and its monomial basis number, namely n_m :

$$n_o = [n_p] = \frac{\sqrt{1+8n_p}-3}{2}, \quad n_p = \langle n_o \rangle = \frac{(n_o+1)(n_o+2)}{2} \quad (50)$$

where $\langle [n] \rangle = n$. Therefore, the dimension of $\text{Im} \mathcal{W}^\Omega$ can be estimated by:

$$\langle \lfloor [n_q] \rfloor - 1 \rangle \leq \dim(\text{Im} \mathcal{W}^\Omega) \leq \langle \lceil [n_q] \rceil - 1 \rangle \quad (51)$$

in which $\lfloor \bullet \rfloor$ and $\lceil \bullet \rceil$ are the floor and ceiling functions, respectively.

For the second term regarding the kernel space of $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$, leading the Lemma.1 listed in Appendix B, the dimension of $Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$ becomes:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) &= \dim(Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}) \\ &= \dim Q_{n_q^\Gamma} - \dim(\ker \mathcal{W}^{\Gamma_q}) \\ &= \dim(\text{Im} \mathcal{W}^{\Gamma_q}) \end{aligned} \quad (52)$$

where $Q_{n_q^\Gamma} \subset Q|_{\Gamma_q}$ is the polynomial space with the identical dimension of trace space $\mathcal{T}^{\Gamma_q}(Q_h)$, i.e. $\dim Q_{n_q^\Gamma} = \dim \mathcal{T}^{\Gamma_q}(Q_h) = 2n_q^\Gamma$. For instance, the detailed monomial basis of $Q_{n_q^\Gamma}$ is listed as follows:

$$Q_{n_q^\Gamma} = \text{span} \left\{ \underbrace{\begin{pmatrix} n_2 \\ -n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ -n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ -n_1 s^2 \end{pmatrix}, \dots}_{\ker \mathcal{W}^{\Gamma_q}}, \underbrace{\begin{pmatrix} n_2 \\ n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ n_1 s^2 \end{pmatrix}, \dots}_{Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}} \right\} \quad (53)$$

134 It is clear that, from Eq. (53), the dimension of second term is evaluated as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) = n_q^{\Gamma} \quad (54)$$

135 With the consideration of Eqs. (47), (51) and (54), the right-hand side of
136 Eq. (44) becomes to:

$$\max(\langle \lfloor [n_q] \rfloor - 1 \rangle, n_q^{\Gamma}) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \leq \langle \lceil [n_q] \rceil - 1 \rangle + n_q^{\Gamma} \quad (55)$$

137 It is noted that the dimension of the trace space $\mathcal{T}^{\Gamma_q}(Q_h)$ is generally iden-
138 tical to the number of DOFs on Γ_q , i.g. for finite element approximation. So
139 generally speaking, $n_q^{\Gamma} \ll \langle \lfloor [n_q] \rfloor - 1 \rangle$. To totally ensure the LBB condition of
140 Eq. (35), considering Eqs. (44), (45) and (55), the n_w should be always less
141 than the low bound of $Q_{n_q} \setminus \ker \mathcal{W}$'s dimension, i.e.:

$$n_w \leq \langle \lfloor [n_q] \rfloor - 1 \rangle \quad (56)$$

142 3.2. Estimator for β_{γ}

143 The behavior of shear strain locking in Eq. (36) is controlled by the coeffi-
144 cient β_{γ} , where the shear strain locking contains two parts as follows:

$$\begin{aligned} & b(\delta \mathbf{q}_h, \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) = (\delta \mathbf{q}_h, \mathcal{Q} \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) \\ & = \int_{\Omega} \delta \mathbf{q} \cdot (\nabla w - \boldsymbol{\varphi}) d\Omega - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} (w - \bar{w}) d\Gamma \\ & = 0 \end{aligned} \quad (57)$$

145 where the product operator herein stands for the tensors. And the operator
146 $\mathcal{Q} : W \times \Phi \rightarrow Q$ is defined individually in the domain Ω and on the boundary
147 Γ_w by:

$$\mathcal{Q} = \begin{cases} \mathcal{Q}^{\Omega} = \{\nabla, -\} & \text{in } \Omega \\ \mathcal{Q}^{\Gamma_w} \mathcal{T}^w, \mathcal{Q}^{\Gamma_w} = -\mathbf{n} & \text{on } \Gamma_w \end{cases} \quad (58)$$

148 in which, in accordance with trace theorem [], $\mathcal{T}^w : W \times \Phi \rightarrow W|_{\Gamma_w}$ is the trace
149 operator mapping displacement $\mathbf{u}$ from domain Ω to boundary Γ_w . Again,
150 introducing the Theorem. 1, the β_{γ} can further be bounded by:

$$\beta_{\gamma} \leq \inf_{U' \subset W_{n_w} \times \Phi_{n_{\varphi}} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u} \sup_{\mathbf{u}_h \in U'} \frac{\|\mathcal{Q} \mathbf{u}\|_Q}{\|\mathbf{u}\|_{W \times \Phi}} + O(h) \quad (59)$$

151 where $\mathcal{I}_h^u : W_{n_w} \times \Phi_{n_{\varphi}} \rightarrow W_h \times \Phi_h$ is the interpolation operator for deflection
152 w and rotation $\boldsymbol{\varphi}$, and W_{n_w} , $\Phi_{n_{\varphi}}$ are the polynomial spaces that have the
153 identical dimensions with W_h , Φ_h respectively, i.e. $\dim W_{n_w} = \dim W_h = n_w$,
154 $\dim \Phi_{n_{\varphi}} = \dim \Phi_h = 2n_{\varphi}$.

155 To maintain the LBB condition of (36), i.e. β_{γ} is a coefficient independent
156 with characterized length h , the dimensions of the relative spaces in Eq. (59)
157 should satisfy the following relationship:

$$\dim(W_{n_w} \times \Phi_{n_{\varphi}} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) \leq \dim(W_{n_w} \times \Phi_{n_{\varphi}} \setminus \ker \mathcal{Q}) \quad (60)$$

158 The left-hand side of Eq. (60) can be represented by the DOFs of approximated
 159 shear force resultant:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) &= \dim(W_h \times \Phi_h \setminus \ker \mathcal{Q}_h) \\
 &= \dim(W_h \times \Phi_h) - \dim(\ker \mathcal{Q}_h) \\
 &= \dim(\text{Im } \mathcal{Q}_h) \\
 &= \dim \mathcal{Q}_h = 2n_q
 \end{aligned} \tag{61}$$

160 On the other hand, with De Morgan's theorem and Grassmann's theorem [],
 161 the right-hand side of Eq. (60) becomes:

$$\begin{aligned}
 &\dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus (\ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w})) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cup \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) + \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &\quad - \dim((W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cap (W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}))
 \end{aligned} \tag{62}$$

162 where the relationship of $\ker \mathcal{Q} = \ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$ is used herein. Similarly,
 163 this dimension of Eq. (62) is consisted of two terms, one relates to the kernel
 164 space of $\mathcal{Q}^\Omega$ in Ω , the other relates to the kernel space of $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$ on Γ_w .

165 The dimension of first term regarding the kernel space of $\mathcal{Q}^\Omega$ can be revealed
 166 by the detailed expression of monomial basis of $W_{n_w} \times \Phi_{n_\varphi}$ as:

$$W_{n_w} \times \Phi_{n_\varphi} = \text{span} \left\{ \overbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} x \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} y \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} x^2 \\ 2x \\ 0 \end{pmatrix}, \begin{pmatrix} xy \\ y \\ x \end{pmatrix}, \begin{pmatrix} y^2 \\ 0 \\ 2y \end{pmatrix}, \dots}^{\ker \mathcal{Q}^\Omega}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ x \end{pmatrix}, \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ y \end{pmatrix}, \dots}_{W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega} \right\} \tag{63}$$

167 Obviously, the monomial basis of deflection w is one order higher than that of
 168 rotation in $\ker \mathcal{Q}^\Omega$. So the dimension of $\ker \mathcal{Q}^\Omega$ can be evaluated by:

$$\dim \ker \mathcal{Q}^\Omega = \min(n_w, n_s) \tag{64}$$

169 with

$$\langle \lfloor n_\varphi \rfloor + 1 \rangle \leq n_s \leq \langle \lceil n_\varphi \rceil + 1 \rangle \tag{65}$$

170 where n_w presents the number of monomial basis for w . n_s stands for the
 171 monomial basis number with one order higher than Φ_{n_φ} 's monomial basis. Then,
 172 the dimension of $W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega$ can be evaluated by:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) &= \dim(W_{n_w} \times \Phi_{n_\varphi}) - \dim(\ker \mathcal{Q}^\Omega) \\
 &= n_w + 2n_\varphi - \min(n_w, n_s)
 \end{aligned} \tag{66}$$

173 To further calculate the dimension of the second term regarding the kernel
 174 space of $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$, we first suppose the trace space $\mathcal{T}^{\Gamma_w}(W_h \times \Phi_h)$ and its
 175 corresponding polynomial space $W_{n_w}^{\Gamma}$, $\dim W_{n_w}^{\Gamma} = \dim \mathcal{T}^{\Gamma_w}(W_h \times \Phi_h) = n_w^{\Gamma}$.
 176 Using the Lemma.1 in Appendix B, we have:

$$\begin{aligned} \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) &= \dim(W_{n_w}^{\Gamma} \setminus \ker \mathcal{Q}^{\Gamma_w}) \\ &= \dim W_{n_w}^{\Gamma} - \underbrace{\dim(\ker \mathcal{Q}^{\Gamma_w})}_0 \\ &= \dim W_{n_w}^{\Gamma} = n_w^{\Gamma} \end{aligned} \quad (67)$$

177 where, it can reveal from the expression of $\mathcal{Q}^{\Gamma_w}$ in Eq.(58) that, the space
 178 $\ker \mathcal{Q}^{\Gamma_w}$ is empty, i.e $\dim(\ker \mathcal{Q}^{\Gamma_w}) = 0$.

179 Consequently, substituting Eqs.(66) and (67) into Eq.(62), the right-hand
 180 side of Eq.(60) can be rewritten as:

$$\begin{aligned} \max(n_w + 2n_\varphi - \min(n_w, n_s), n_w^{\Gamma}) &\leq \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\ &\leq n_w + 2n_\varphi - \min(n_w, n_s) + n_w^{\Gamma} \end{aligned} \quad (68)$$

181 It is clear that $n_w^{\Gamma} \ll n_w + 2n_\varphi - \min(n_w, n_s)$ FIXME

$$n_q \leq n_\varphi \quad (69)$$

$$n_w^{\text{opt}} = \langle \lfloor n_q \rfloor - 1 \rangle, \quad n_q^{\text{opt}} = n_\varphi \quad (70)$$

183 **Remark 3.** It should be noted that, the proposed estimator is considered for
 184 the entire domain and all the nodal discretization should be so well-posed that
 185 the monomial basis number of equivalent polynomial space is equal to the degree
 186 of freedoms for approximation space, i.g $\dim W_{n_w} = \dim W_h = n_w$. While
 187 a local domain contains a too dense nodal distribution than others, the nodal
 188 discretization satisfied the estimator will still meet the locking issue in local.

189 **Remark 4.** The shear force locking will influence the shear strain locking.

190 4. Shear-locking-free FE-Meshfree formulation

191 4.1. Reproducing kernel meshfree approximation

$$\Psi_I(\mathbf{x}) = \mathbf{c}^T(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (71)$$

$$\phi(\mathbf{x}_I - \mathbf{x}) = \phi(r_{1I}) \phi(r_{2I}), \quad r_{iI} = \frac{|x_{iI} - x_i|}{s_i} \quad (72)$$

$$\phi(r) = \frac{1}{3!} \begin{cases} (2-2r)^3 - 4(1-2r)^3 & r \leq \frac{1}{2} \\ (2-2r)^3 & \frac{1}{2} < r \leq 1 \\ 0 & r > 1 \end{cases} \quad (73)$$

194

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I) = \mathbf{p}(\mathbf{x}) \quad (74)$$

195 or a shifted form

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) = \mathbf{p}(\mathbf{0}) \quad (75)$$

196

$$\mathbf{c}(\mathbf{x}_I - \mathbf{x}) = \mathbf{A}^{-1}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{0}) \quad (76)$$

197 with

$$\mathbf{A}(\mathbf{x}_I - \mathbf{x}) = \sum_{I=1}^n \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}^T(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (77)$$

198

$$\Psi_I(\mathbf{x}) = \mathbf{p}^T(\mathbf{0}) \mathbf{A}^{-1} \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (78)$$

199 *4.2. Eigenvalue test for checking LBB coefficients(should be fixed)*

200 To verify the optimal constraint ratios proposed in Eq. (70), the eigenvalue
 201 test of LBB coefficients [4] in a square domain, shown in Figure.??, is intro-
 202 duced herein, where all the boundaries of this domain are clamped. In order
 203 to adjust the constraint ratio more flexibly, the reproducing kernel meshfree
 204 approximation are used to discretize the deflection w and rotation φ , yields:

$$N_I^w := \Psi_I, N_K^\varphi := \Psi_K^\varphi, N_R^q := N_R \quad (79)$$

205 while, as shown in Figure.??, the meshfree nodes for w and φ are different.

206 To meet the variational consistency of the weak formulation with meshfree
 207 approximation, the reproducing kernel gradient smoothing integration frame-
 208 work (RKGSi) [5] is used herein. Accordingly, the Mindlin plate version of
 209 RKGSi framework is presented in Appendix C.

210 *4.3. Shear-locking-free formulations with optimal constraint ratio*

$$N_I^w := \Psi_I, N_K^\varphi := N_K, N_R^q := N_R \quad (80)$$

211 FIXME

212 **Remark 5.** *The discretization of Eq.*213 **5. Numerical examples**214 *5.1. Square plate subjected non-uniform transverse load*

215 Consider a square plate with the clamped boundary, as showing in Figure.
 216 , subjecting a non-uniform transverse load $\bar{p}$:

$$\bar{p} = \frac{Eh^3}{12(1-\nu^2)} \left(\begin{array}{l} 12y(y-1)(5x^2-5x+1)(2y^2(y-1)^2+x(x-1)(5y^2-5y+1)) \\ +12x(x-1)(5y^2-5y+1)(2x^2(x-1)^2+y(y-1)(5x^2-5x+1)) \end{array} \right) \quad (81)$$

217 In this circumstance, the exact solution of the deflection and rotation are listed
 218 as follows:

$$w = \frac{1}{3}x^3(x-1)^3y^3(y-1)^3 - \frac{2h^2}{5(1-\nu)} \left(\frac{y^3(y-1)^3x(x-1)(5x^2-5x+1)+}{x^3(x-1)^3y(y-1)(5y^2-5y+1)} \right) \quad (82)$$

$$\begin{cases} \varphi_1 = y^3(y-1)^3x^2(x-1)^2(2x-1) \\ \varphi_2 = x^3(x-1)^3y^2(y-1)^2(2y-1) \end{cases} \quad (83)$$

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224 Appendix A. Complete-polynomial-based estimator for LBB coeffi- 225 cient

226 This appendix presents the detailed derivation of the LBB coefficient esti-
227 mator used in this work. This estimator provides a quantitative measure of
228 the stability of mixed finite element formulations just by counting the degree
229 of freedoms (DOFs) of the approximation spaces. The main route of derivation
230 follows the path proposed in Ref. [2]. Unlike the consideration of volumetric
231 locking in [2], here we focus on the general LBB condition for any locking is-
232 sue. Consider the following constraint servicing to a saddle-point problem: find
233 $u \in U$ such that

$$b(p, u) = g(p), \quad \forall p \in P \quad (\text{A.1})$$

234 where $b : U \times P \rightarrow \mathbb{R}$, $g : P \rightarrow \mathbb{R}$ are bilinear and linear forms, respectively. For
235 an any fixed $p \in P$, there exists a linear operator $\mathcal{P} : U \rightarrow P$ satisfying:

$$b(p, u) = (p, \mathcal{P}u) \quad (\text{A.2})$$

236 where $(\bullet, \bullet) : P \times P \rightarrow \mathbb{R}$ is the inner product in P .

237 By introducing the discrete spaces $U_h \subset U$ and $P_h \subset P$, the constraint of
238 Eq. (A.1) is approximated as: find $u_h \in U_h$ such that

$$b(p_h, u_h) = g(p_h), \quad \forall p_h \in P_h \quad (\text{A.3})$$

239 and $\mathcal{P}$'s orthogonal projection operator regarding to P_h , namely $\mathcal{P}_h : U_h \rightarrow P_h$,
240 is defined by

$$(p_h, \mathcal{P}u_h - \mathcal{P}_h u_h) = 0 \Rightarrow (p_h, \mathcal{P}u_h) = (p_h, \mathcal{P}_h u_h), \quad \forall p_h \in P_h \quad (\text{A.4})$$

241 To obtain the satisfaction solution of saddle-point problems with constraint
242 of Eq. (A.3), the well-known LBB condition is required, i.e., there exists a
243 positive constant $\beta > 0$, namely LBB coefficient, such that

$$\inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|b(p_h, u_h)|}{\|p\|_P \|u\|_U} \geq \beta \quad (\text{A.5})$$

244 where $\|\bullet\|_U$ and $\|\bullet\|_P$ are the norms in spaces U and P , respectively. To
245 further estimate the LBB coefficient β by the degree of freedom of U_h and P_h ,
246 i.e. $n_u = \dim U_h$ and $n_p = \dim P_h$, we introduce the following theorem.

247 **Theorem 1.** (Completed-polynomial-based estimator for LBB coefficient) Sup-
248 pose that $\mathcal{P} : U \rightarrow P$ is the linear operator representing the bilinear form b
249 in LBB condition and $\mathcal{P}_h : U_h \rightarrow P_h$ is the corresponding orthogonal projec-
250 tion operator regarding to P_h , $U_{n_u} \subset U$ is a complete polynomial space with
251 $\dim U_{n_u} = n_u$. The LBB coefficient β can be estimated by:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}u\|_P}{\|u\|_U} + O(h) \quad (\text{A.6})$$

252 where $\mathcal{I}_h : U_{n_u} \rightarrow U_h$ is the interpolation operator with the characteristic size
253 of the discretization h .

254 PROOF. Firstly, with the definition of $\mathcal{P}$ and $\mathcal{P}_h$ in Eqs. (A.2) and (A.4) re-
 255 spectively, the LBB condition can be rewritten as:

$$\begin{aligned} \beta &\leq \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}u_h)|}{\|p_h\|_P \|u_h\|_U} = \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \\ &\leq \inf_{p_h \in \text{Im} \mathcal{P}_h} \sup_{u \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \end{aligned} \quad (\text{A.7})$$

256 FIXME: the definition of image space can be canceled. where $\text{Im} \mathcal{P}_h$ is the
 257 image space of operator $\mathcal{P}_h$, i.e. $\text{Im} \mathcal{P}_h := \{p_h \in P_h \mid \exists u_h \in U_h, \mathcal{P}_h u_h = p_h\}$,
 258 $\text{Im} \mathcal{P}_h \subset P_h$. Since both p_h and $\mathcal{P}_h u_h$ belong to $\text{Im} \mathcal{P}_h$, using Cauchy-Schwarz
 259 inequality, we can further obtain that:

$$|(\mathcal{P}_h u_h, p_h)| \leq \|\mathcal{P}_h u_h\|_P \|p_h\|_P \quad (\text{A.8})$$

260 For a given p_h , with introduction of a space $U' := \{u_h \in U_h \mid \mathcal{P}_h u_h = p_h\}$
 261 regarding to p_h , the equality in Eq. (A.7) holds when $p_h = \mathcal{P}_h u_h$, yields:

$$|(\mathcal{P}_h u_h, p_h)| = \|\mathcal{P}_h u_h\|_P \|p_h\|_P, \quad \forall u_h \in U' \quad (\text{A.9})$$

262 and, obviously, $U' \subseteq U_h \setminus \ker \mathcal{P}_h$, $\ker \mathcal{P}_h$ is the kernel space of operator $\mathcal{P}_h$, i.e.
 263 $\ker \mathcal{P}_h := \{u_h \in U_h \mid \mathcal{P}_h u_h = 0\}$. Substituting Eq. (A.9) into Eq. (A.7) leads
 264 to:

$$\beta \leq \inf_{U' \subset U_h \setminus \ker \mathcal{P}_h} \sup_{u_h \in U'} \frac{\|\mathcal{P}_h u_h\|_P}{\|u_h\|_U} \quad (\text{A.10})$$

265 Secondly, the complete polynomial space $V_{n_u} \subset U$ with identical dimension
 266 to U_h is introduced herein to further estimate the β , where n_u is the dimension
 267 of U_h , i.e. $n_u = \dim U_h = \dim V_{n_u}$. At this circumstance, for any $u \in U_{n_u}$,
 268 there exists an interpolation operator $\mathcal{I}_h : U \rightarrow U_h$ such that $u_h = \mathcal{I}_h u$. Thus,
 269 Eq. (A.10) can be rewritten as:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \quad (\text{A.11})$$

270 In accordance with the triangular inequality [6] and the relationship of Eq.
 271 (A.4), we have:

$$\begin{aligned} \|\mathcal{P}_h \mathcal{I}_h u\|_P &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P}_h \mathcal{I}_h u, p_h)|}{\|p_h\|_P} = \sup_{p_h \in P_h} \frac{|(\mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P} u, p_h)| + |(\mathcal{P} u - \mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &= \|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P \end{aligned} \quad (\text{A.12})$$

272 where the last term is the interpolation error of space U_h with respect to operator
 273 $\mathcal{P}$. Generally, particularly for finite element approximations, the interpolation
 274 error is proportional to the characteristic size of the discretization h [6]:

$$\|\mathcal{P}(u - \mathcal{I}_h u)\|_P \leq Ch^k |u|_{H_k} \quad (\text{A.13})$$

275 in which C is a positive constant independent of h , k is the convergence order
 276 related to the operators $\mathcal{I}_h$ and $\mathcal{P}$. With the combination of Eqs. (A.11), (A.12)
 277 and (A.13), the final estimator of LBB coefficient β in Eq. (1) can be proved:

$$\begin{aligned}
 \beta &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P}{\|u\|_U} + O(h)
 \end{aligned} \tag{A.14}$$

278 FIXME:use the remark env It should be noted that Theorem 1 inherently
 279 embeds the numbers of DOFs of approximation spaces U_h and P_h , while $n_u =$
 280 $\dim U_h$ and $n_p = \dim \text{Im} \mathcal{P}_h = \dim(U_h \setminus \ker \mathcal{P}_h \mathcal{I}_h)$, thus it is easy to justify the
 281 LBB condition by just counting the DOFs of approximation spaces.

282 **Appendix B. Preliminaries for functional analysis**

283 In the interest of conciseness, kernel space

$$\ker \mathcal{P} := \{u \in U \mid \mathcal{P}u = 0\} \quad (\text{B.1})$$

284 image space

$$\text{Im} \mathcal{P} := \{p \in P \mid \exists u \in U, p = \mathcal{P}u\} \quad (\text{B.2})$$

285 It is clear that $\ker \mathcal{P} \subset U$ and $\text{Im} \mathcal{P} \in P$.

286 **FIXME:** Trace's theorem De Morgan's theorem Grassmann's theorem Rank-
287 nullity theorem

288 **Lemma 1.** *Suppose that $\mathcal{T} : U \rightarrow U|_{\Gamma}$ is the trace operator of U , $U|_{\Gamma}$ is U 's
289 trace space on the boundary Γ . Let $\mathcal{L}$ is an operator acting on $U|_{\Gamma}$, then we
290 have:*

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.3})$$

291 **PROOF.** In accordance with rank-nullity theorem, we have the following rela-
292 tionship:

$$\dim U = \dim(\text{Im} \mathcal{L}\mathcal{T}) + \dim(\ker \mathcal{L}\mathcal{T}) \quad (\text{B.4})$$

293 or

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim U - \dim(\ker \mathcal{L}\mathcal{T}) = \dim(\text{Im} \mathcal{L}\mathcal{T}) \quad (\text{B.5})$$

294 in which, according to the definition of the image space $\text{Im} \mathcal{L}\mathcal{T}$, it is clear that:

$$\text{Im} \mathcal{L}\mathcal{T} = \{\mathcal{L}\mathcal{T}u \mid u \in U\} = \{\mathcal{L}(\mathcal{T}u) \mid \mathcal{T}u \in U|_{\Gamma}\} = \text{Im} \mathcal{L} \quad (\text{B.6})$$

295 Subsequently, the dimension of image space for operator $\mathcal{L}$, i.e. $\dim(\text{Im} \mathcal{L})$,
296 can be evaluated by introducing rank-nullity theorem again as follows:

$$\dim U|_{\Gamma} = \dim(\text{Im} \mathcal{L}) + \dim(\ker \mathcal{L}) \quad (\text{B.7})$$

297 or

$$\dim(\text{Im} \mathcal{L}) = \dim U|_{\Gamma} - \dim(\ker \mathcal{L}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.8})$$

298 With the combination of Eqs. (B.5), (B.6) and (B.8), the relationship of Eq.
299 (B.3) can be proofed.

300 **Appendix C. Reproducing kernel smoothed gradient method for Mindlin**
 301 **plate formulation**

$$\begin{aligned}
 \Pi(w, \boldsymbol{\varphi}, \mathbf{q}, \mathbf{m}) = & - \int_{\Omega} \frac{1}{2} \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Omega} \frac{1}{2} \frac{\mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega \\
 & - \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
 & - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot (\mathbf{m} \cdot \mathbf{n} - \bar{\mathbf{m}}_n) d\Gamma - \int_{\Omega} \boldsymbol{\varphi} \cdot (\nabla \cdot \mathbf{m} - \mathbf{q}) d\Omega \\
 & + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{\mathbf{q}}_n) d\Gamma - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega
 \end{aligned} \tag{C.1}$$

$$\begin{aligned}
 a_m(\delta \mathbf{m}, \mathbf{m}) + b_m(\delta \mathbf{m}, \boldsymbol{\varphi}) &= f_m(\delta \mathbf{m}), \quad \forall \delta \mathbf{m} \in M \\
 a_q(\delta \mathbf{q}, \mathbf{q}) + b_q(\delta \mathbf{q}, \mathbf{u}) &= f_q(\delta \mathbf{q}), \quad \forall \delta \mathbf{q} \in Q \\
 b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\
 b_m(\delta \boldsymbol{\varphi}, \mathbf{m}) + b_{\varphi}(\delta \boldsymbol{\varphi}, \mathbf{q}) &= f_{\varphi}(\delta \boldsymbol{\varphi}), \quad \forall \delta \boldsymbol{\varphi} \in \Phi
 \end{aligned} \tag{C.2}$$

$$a_m(\delta \mathbf{m}, \mathbf{m}) = - \int_{\Omega} \delta \mathbf{m} : \frac{12}{t^3} \mathbf{D}^{-1} : \mathbf{m} d\Omega \tag{C.3}$$

$$b_m(\mathbf{m}, \boldsymbol{\varphi}) = - \int_{\Gamma_m} \mathbf{n} \cdot \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega \tag{C.4}$$

$$f_m(\boldsymbol{\varphi}, \mathbf{q}) = \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \tag{C.5}$$

$$\tag{C.6}$$

$$\begin{aligned}
 - \int_{\Omega} \delta \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Gamma_m} \mathbf{n} \cdot \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega &= \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \delta \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \\
 - \int_{\Omega} \frac{\delta \mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega + \int_{\Gamma_q} \delta \mathbf{q} \cdot \mathbf{n} w d\Gamma - \int_{\Omega} \nabla \cdot \delta \mathbf{q} w d\Omega + \int_{\Omega} \delta \mathbf{q} \cdot \boldsymbol{\varphi} d\Omega &= - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma
 \end{aligned} \tag{C.7}$$

$$\begin{aligned}
 - \int_{\Omega} \frac{\delta \mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega + \int_{\Gamma_q} \delta \mathbf{q} \cdot \mathbf{n} w d\Gamma - \int_{\Omega} \nabla \cdot \delta \mathbf{q} w d\Omega + \int_{\Omega} \delta \mathbf{q} \cdot \boldsymbol{\varphi} d\Omega &= - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
 \int_{\Gamma_q} \delta w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta w \nabla \cdot \mathbf{q} d\Omega &= \int_{\Gamma_q} \delta w \bar{\mathbf{q}}_n d\Gamma - \int_{\Omega} \delta w \bar{p} d\Omega
 \end{aligned} \tag{C.8}$$

$$\int_{\Gamma_q} \delta w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta w \nabla \cdot \mathbf{q} d\Omega = \int_{\Gamma_q} \delta w \bar{\mathbf{q}}_n d\Gamma - \int_{\Omega} \delta w \bar{p} d\Omega \tag{C.9}$$

$$\begin{aligned}
 - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \nabla d\Omega + \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega &= - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \\
 - \int_{\Omega_C} \delta m_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} m_{\gamma\eta} d\Omega - \int_{\Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma + \int_{\Omega_C} \delta M_{\alpha\beta, \beta} \varphi_{\alpha} d\Omega + \int_{\Gamma_{\varphi} \cap \Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma &= \int_{\Gamma_{\varphi} \cap \Gamma_C} M_{\alpha\beta} n_{\beta} \bar{\varphi}_{\alpha} d\Gamma
 \end{aligned} \tag{C.10}$$

$$\begin{aligned}
 - \int_{\Omega_C} \delta m_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} m_{\gamma\eta} d\Omega - \int_{\Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma + \int_{\Omega_C} \delta M_{\alpha\beta, \beta} \varphi_{\alpha} d\Omega + \int_{\Gamma_{\varphi} \cap \Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma &= \int_{\Gamma_{\varphi} \cap \Gamma_C} M_{\alpha\beta} n_{\beta} \bar{\varphi}_{\alpha} d\Gamma \\
 &\tag{C.11}
 \end{aligned}$$

$$m_{\alpha\beta h}(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{a}_{\alpha\beta C} \quad \text{in } \Omega_C \quad (\text{C.12})$$

$$\mathbf{a}_{\alpha\beta C} = D_{\alpha\beta\eta\gamma} \mathbf{G}_C^{-1} \left(\sum_{K=1}^{n_\varphi} (\tilde{\mathbf{g}}_{\eta K} + \hat{\mathbf{g}}_{\eta K}) \varphi_{\gamma K} + \bar{\mathbf{g}}_{\gamma\eta} \right) \quad (\text{C.13})$$

$$\tilde{\mathbf{g}}_{\alpha K} = - \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\alpha d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\alpha} N_K^\varphi d\Omega \quad (\text{C.14})$$

$$\bar{\mathbf{g}}_{\alpha K} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\alpha d\Gamma \quad (\text{C.15})$$

$$\hat{\mathbf{g}}_{\alpha\beta} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} n_\beta \bar{\varphi}_\alpha d\Gamma \quad (\text{C.16})$$

$$(\text{C.17})$$

$$\mathbf{m}_h = \tilde{\mathbf{m}}_h + \hat{\mathbf{m}}_h + \bar{\mathbf{m}}_h \quad (\text{C.18})$$

$$\mathbf{m}(\varphi_h) = \mathbf{D} : (\tilde{\boldsymbol{\kappa}}(\varphi_h) + \hat{\boldsymbol{\kappa}}_h + \bar{\boldsymbol{\kappa}}_{\gamma\eta}) \quad (\text{C.19})$$

$$\tilde{m}_{\alpha\beta h} = -D_{\alpha\beta\eta\gamma} (\tilde{\kappa}_{\gamma\eta h} + \bar{\kappa}_{\gamma\eta} + \hat{\kappa}_{\gamma\eta}) \quad (\text{C.20})$$

with

$$\tilde{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left(\tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \tilde{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.21})$$

$$\bar{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left(\bar{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \bar{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.22})$$

$$\hat{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\mathbf{G}_C^{-1} \hat{\mathbf{g}}_{\alpha\beta} \quad (\text{C.23})$$

$$\tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\alpha K} \quad (\text{C.24})$$

$$\bar{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\alpha K} \quad (\text{C.25})$$

$$\begin{aligned} & b_m(\delta\varphi_h, \mathbf{m}_h) \\ &= - \int_{\Gamma_m} \delta\varphi_h \cdot \mathbf{m}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \mathbf{m}_h \cdot \nabla d\Omega \\ &= - \int_{\Gamma} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma + \int_{\Omega} \delta m_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_\varphi} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left(- \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\beta d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^\varphi d\Omega + \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\beta d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} ((\tilde{\mathbf{g}}_{\eta L} + \bar{\mathbf{g}}_{\eta L}) \varphi_{\gamma L} + \hat{\mathbf{g}}_{\gamma\eta}) \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left(\int_{\Omega_C} \tilde{\mathbf{g}}_{\beta K}^T \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \right) \varphi_{\gamma L} \end{aligned} \quad (\text{C.26})$$

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$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \underbrace{\int_{\Gamma} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega}_{\int_{\Omega} \delta\tilde{\kappa} : \frac{t^3}{12} \mathbf{D} : \tilde{\kappa} d\Omega} + \int_{\Gamma_{\varphi}} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma \quad (\text{C.27})
\end{aligned}$$

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$$b_m(\delta\varphi_h, \hat{\mathbf{m}}_h) = \tilde{b}_{\varphi}(\varphi_h, \tilde{\mathbf{m}}(\delta\varphi_h)) + \tilde{b}_{\varphi}(\delta\varphi_h, \hat{\mathbf{m}}(\varphi_h)) \quad (\text{C.28})$$

317

$$b_m(\delta\varphi_h, \bar{\mathbf{m}}_h) = \hat{f} + \bar{f} \quad (\text{C.29})$$

318

$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \int_{\Gamma} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma + \int_{\Omega} \delta\tilde{m}_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_{\varphi}} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} \left(- \int_{\Gamma_C} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^{\varphi} d\Omega + \int_{\Gamma_C \cap \Gamma_{\varphi}} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} (\tilde{\mathbf{g}}_{\beta K} + \hat{\mathbf{g}}_{\beta K}) \mathbf{G}_C^{-1} \int_{\Omega_C} \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{p}^T d\Omega \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \left(\underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\tilde{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&+ \sum_{C=1}^{n_e} \left(\underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\bar{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&= a_{\varphi}(\tilde{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) + a_{\varphi}(\hat{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) \quad (\text{C.30})
\end{aligned}$$

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